

R_q : Decompositions, Automorphisms, and Embeddings

Lecture 4 of 6 — Foundations

SEC 595 — Encrypted Computing

Where we are going

Lecture 3 built $R_q = \mathbb{Z}_q[X]/(X^n + 1)$ and showed how to multiply in it quickly.

That is enough to *state* BGV, BFV and CKKS. It is not enough to *run* them.

Between the definition of a scheme and a working implementation sit seven pieces of machinery. This lecture supplies all seven.

The seven tools, and how each is presented

- 1 RNS and the modulus chain
- 2 Modulus switching
- 3 Gadget decomposition
- 4 $R_q[S]$ and relinearisation
- 5 Batching: SIMD slots
- 6 Galois automorphisms and rotation
- 7 The canonical embedding and CKKS encoding

Each one gets: **the property, where it is used**, and a **worked example**.
The examples are the explanation, and we work them here together.

- n a power of two, $R = \mathbb{Z}[X]/(X^n + 1)$, $R_q = R/qR$;
- centered representatives and the norm $\|\cdot\|_\infty$ of Lecture 1;
- the bound $\|ab\|_\infty \leq n \|a\|_\infty \|b\|_\infty$ of Lecture 3.

Worked examples use $n = 4$, or a single coefficient where the identity being checked does not involve n .

The property — the residue number system

Let $Q = q_0 q_1 \cdots q_L$ with the q_i pairwise coprime. The **RNS representation** of $a \in R_Q$ is

$$([a]_{q_0}, [a]_{q_1}, \dots, [a]_{q_L}),$$

its coefficientwise reductions, and

$$R_Q = \mathbb{Z}_Q[X]/(X^n + 1) \cong R_{q_0} \times R_{q_1} \times \cdots \times R_{q_L},$$

an isomorphism for both $+$ and $\times$.

The property — the modulus chain

Writing

$$q_\ell = p_0 p_1 \cdots p_\ell, \quad \ell = 0, 1, \dots, L,$$

gives the **modulus chain**. A ciphertext modulo q_ℓ is at **level** ℓ .

Dropping from q_ℓ to $q_{\ell-1}$ deletes one component of the tuple.

Every practical BGV, BFV and CKKS library stores a ring element this way: an $(L + 1) \times n$ array of machine words, the q_i primes just under 2^{60} , every operation componentwise.

- BGV, Section 2 takes the chain over verbatim, with 17, 41, 97 as its own running example;
- CKKS, Section 7 counts levels the same way;
- L is the multiplicative depth the parameters support, and no multiprecision division occurs anywhere.

Example — the primes and the idempotents

Take $n = 4$ and

$$p_0 = 17, \quad p_1 = 41, \quad p_2 = 97.$$

Compute $Q = p_0 p_1 p_2$. Check that each $p_i \equiv 1 \pmod{8}$, and say what that congruence buys from Lecture 3.

Then compute the three CRT idempotents e_0, e_1, e_2 by the method of Lecture 1, and verify $e_0 + e_1 + e_2 \equiv 1 \pmod{Q}$.

Example — the round trip, and the chain

Same primes, $Q = p_0 p_1 p_2$.

Map the coefficient $a = 12345$ into RNS form, then reconstruct it from its residues using the idempotents.

Finally write down the three-rung chain q_2, q_1, q_0 on these primes.

The property — correctness of the switch

Let ct be a ciphertext modulo q with $ct[SK] = m + tv + qk$ over R , let $q' \mid q$, and let ct' be the integer polynomial closest to $(q'/q) ct$ subject to $ct' \equiv ct \pmod{t}$.

Proposition 1. If $q \equiv q' \pmod{t}$, then

$$[ct'[SK]]_{q'} = m + tv'$$

with

$$\|v'\|_{\infty} \leq \frac{q'}{q} \|v\|_{\infty} + \frac{1}{2}(1 + n \|SK\|_{\infty}) + \frac{\|m\|_{\infty}}{t},$$

provided the right-hand side is below $q'/(2t)$.

- BGV, Section 10 calls this after every multiplication. It is what makes the scheme *levelled* rather than merely somewhat homomorphic.
- Each call spends one prime of the chain of Section 1.
- CKKS, Section 6 runs the same formula $\lfloor (q'/q) ct \rfloor$ with no congruence condition and calls it `RESCALE`, using it to restore the scale $\Delta^2 \mapsto \Delta$ after a multiplication.

Example — setting up the ciphertext

Take $t = 8$, $q = q_2 = 67609$, $q' = q_1 = 697$, secret $SK = 3$. Track a single coefficient, since the switch is coefficientwise.

Let $m = 5$, $v = 2$, and $c_1 = 41234$.

Find c_0 so that $[ct[SK]]_q = m + tv$, and check that decryption returns m .

Example — performing the switch

Same data: $t = 8$, $q = 67609 \rightarrow q' = 697$, $SK = 3$, and the pair (c_0, c_1) just built.

- a. Switch both components: round $(q'/q)c$ to the nearest integer congruent to c modulo 8.
- b. Decrypt at q' and read off the new noise v' .
- c. Compare with the bound of Proposition 1 at $n = 1$.

The property — base- B digits and their bound

Fix a base $B \geq 2$, set $\ell = \lceil \log_B q \rceil$, and let

$$g = (1, B, B^2, \dots, B^{\ell-1})$$

be the **gadget vector**. For $a \in R_q$ let $G^{-1}(a) = (a_0, \dots, a_{\ell-1})$ be the balanced base- B digits of its centered representative.

Proposition 2.

$$\langle g, G^{-1}(a) \rangle = \sum_{i=0}^{\ell-1} B^i a_i = a \text{ in } R_q, \quad \|G^{-1}(a)\|_{\infty} \leq B/2,$$

the second bound **uniformly in a** .

Whenever a scheme must multiply a ring element by something carrying an error term, it multiplies by the *digits* instead.

- relinearisation in Section 4, key switching in Section 6;
- BGV, Section 9 and BFV, Section 10 publish one evaluation key per digit position on exactly this basis;
- GSW, Section 2 is the gadget matrix itself: it restates Proposition 2 as *flatness* and cites it as “Lecture 4, Proposition 2”.

Example — a base- B decomposition

Take $q = 64$ and $B = 4$.

- a. Compute ℓ and write down g .
- b. Decompose $a = 45$: take the centered representative and extract the balanced base-4 digits.
- c. Verify $\langle g, G^{-1}(45) \rangle = 45$ in $\mathbb{Z}_{64}$, and compare $\|G^{-1}(45)\|_{\infty}$ with $B/2$ and with $\|45\|_{\infty}$.
- d. Repeat at $B = 2$.

The property — the ciphertext as a polynomial in the secret

Identify

$$ct = (c_0, c_1) \quad \text{with} \quad ct[S] = c_0 + c_1 S \in R_q[S],$$

so that decryption is evaluation at $S = SK$.

A product of two ciphertexts is then the degree-2 polynomial

$$ct^*[S] = c_1^* + c_2^* S + c_3^* S^2.$$

The property — relinearisation

Let $(c_{3,0}^*, \dots, c_{3,\ell-1}^*) = G^{-1}(c_3^*)$ and

$$EK^{(i)} = (-(a_i \cdot SK + e_i) + B^i SK^2, a_i), \quad i = 0, \dots, \ell - 1.$$

Set

$$\hat{c}_1 = c_1^* + \sum_i EK_1^{(i)} c_{3,i}^*, \quad \hat{c}_2 = c_2^* + \sum_i EK_2^{(i)} c_{3,i}^*.$$

Proposition 3. $\hat{ct} = (\hat{c}_1, \hat{c}_2)$ is a degree-1 ciphertext under the same secret key, with

$$\hat{ct}[SK] = ct^*[SK] - \sum_i c_{3,i}^* e_i.$$

The $EK^{(i)}$ are published with the public key as the **evaluation key**.

- BFV, Section 9 and BGV, Section 8 produce the three-component ct^* ;
- BFV, Section 10, BGV, Section 9 and CKKS, Section 5 relinearise exactly as above, and all three cite Lecture 4, Section 4 for the reading $ct = c_0 + c_1 S$ rather than redo it.

Nothing here mentions a plaintext modulus or a scale — which is why the three schemes can share it.

Example — a relinearisation in full

Take $q = 64$, $B = 4$, $\ell = 3$, $SK = 3$, and $n = 1$ so every product is an integer product. Let

$$ct^* = (c_1^*, c_2^*, c_3^*) = (5, 12, 45), \quad a_i = (7, 23, 51), \quad e_i = (1, -1, 1).$$

- a. Evaluate $ct^*[SK]$ modulo 64.
- b. Build the three evaluation keys and check $EK_1^{(i)} + a_i SK = B^i SK^2 - e_i$.
- c. Contract against $G^{-1}(45)$ to get $\hat{c}_1, \hat{c}_2$.
- d. Confirm $\hat{c}t[SK] = ct^*[SK] - \sum_i c_{3,i}^* e_i$, and state the added error.

Sections 1–4 complete.

So far: RNS and the modulus chain, modulus switching and `RESCALE`, gadget decomposition, and the ciphertext read as a polynomial in the secret.

Session 9 resumes at *Batching: SIMD slots*.

The property — R_t splits into n slots

Let t be prime with $t \equiv 1 \pmod{2n}$ and let $\zeta \in \mathbb{F}_t^\times$ have order $2n$.

Theorem 1.

$$R_t = \mathbb{F}_t[X]/(X^n + 1) \cong \prod_{j \in \mathbb{Z}_{2n}^\times} \mathbb{F}_t[X]/(X - \zeta^j) \cong \mathbb{F}_t^n,$$

by $a \mapsto (a(\zeta^j))_{j \in \mathbb{Z}_{2n}^\times}$, respecting $+$ and $\times$ componentwise. The n components are the **slots**.

The property — encode and decode

$$\text{Decode}(\mathbf{a}) = (a(\zeta), a(\zeta^3), \dots, a(\zeta^{2^n-1})), \quad \text{Encode} = \text{Decode}^{-1}.$$

Encode is the inverse negacyclic NTT of Lecture 3, so it is free.

One homomorphic addition performs n independent additions, and one homomorphic multiplication n independent multiplications. This is **batching**, and it is what makes the amortised cost per data item tolerable.

- BFV, Section 5 selects the plaintext modulus by this congruence for exactly that reason;
- GSW, Section 1 cites it as the packing GSW gives up;
- CKKS, Section 3 is the same construction over $\mathbb{C}$ — our Section 7.

Example — the four evaluation points

Take $n = 4$ and $t = 17$, so $17 \equiv 1 \pmod{8}$.

From Lecture 2, 3 is a primitive root of $\mathbb{F}_{17}$. Find an element ζ of order 8 and list its powers.

Which four are the primitive 8-th roots? Check that each satisfies $x^4 = -1$, and that they are exactly the roots of $X^4 + 1$ over $\mathbb{F}_{17}$.

Example — a round trip, and one multiplication

Same $n = 4$, $t = 17$, nodes $(\zeta, \zeta^3, \zeta^5, \zeta^7)$.

- a. Encode the slot vector $z = (3, 1, 4, 1)$, then decode your answer to check it.
- b. Encode $z' = (2, 0, 5, 7)$.
- c. Multiply the two encodings in R_{17} , using $X^4 = -1$, and read off the slots of the product.

The property — the automorphisms σ_k

For odd k , let

$$\sigma_k : R \rightarrow R, \quad a(X) \mapsto a(X^k) \bmod (X^n + 1).$$

Each σ_k is a ring automorphism, $\sigma_k = \sigma_{k'}$ exactly when $k \equiv k' \pmod{2n}$, and they form the group

$$\mathbb{Z}_{2n}^\times = \langle 5 \rangle \times \langle -1 \rangle, \quad 5 \text{ of order } n/2.$$

The property — what they do to slots

On the slots of Theorem 1,

$$(\sigma_k(a))(\zeta^j) = a(\zeta^{jk}),$$

so σ_k permutes slots by $j \mapsto jk$ in $\mathbb{Z}_{2n}^\times$.

This is how data moves between slots — something $+$ and $\times$ cannot do.

Applied to a ciphertext, σ_k is an index permutation with sign flips and costs no multiplications. But the result decrypts under $\sigma_k(SK)$, not SK .

- So every rotation is followed by a **key switch**: Proposition 3 with SK^2 replaced by $\sigma_k(SK)$, one switching key per rotation amount used;
- CKKS, Section 3 rotates its slots with σ_5 and conjugates them with σ_{-1} ;
- BGV, Section 9 notes that rotation and relinearisation are the same machinery; CKKS bootstrapping is built almost entirely from sums of rotations.

Example — σ_5 acting on slots

Take $n = 4$, $t = 17$, nodes $(9, 15, 8, 2)$, and $a(X) = 1 + 2X + 3X^2 + 4X^3$.

- a. Identify $\mathbb{Z}_8^\times$, the subgroups $\langle 5 \rangle$ and $\langle -1 \rangle$, and check that they generate it.
- b. Compute the four slots of a .
- c. Compute $\sigma_5(a)$ in R_{17} and its slots.
- d. Verify that the rearrangement is the permutation $j \mapsto 5j$ on $\mathbb{Z}_8^\times$.

The property — the canonical embedding

Let $K = \mathbb{Q}[X]/(X^n + 1)$ and $\zeta = \exp(\pi i/n)$. The **canonical embedding** is

$$\sigma : K \longrightarrow \mathbb{C}^n, \quad \sigma(a) = (a(\zeta), a(\zeta^3), \dots, a(\zeta^{2n-1})),$$

an injective ring homomorphism.

Keeping one coordinate from each conjugate pair — the subgroup $\langle 5 \rangle \subset \mathbb{Z}_{2n}^\times$ — gives the $\mathbb{R}$ -linear multiplicative bijection

$$\tau : K \rightarrow \mathbb{C}^{n/2}, \quad \tau(a) = (a(\zeta), a(\zeta^5), a(\zeta^{25}), \dots).$$

For a scale factor Δ ,

$$\text{Encode}(z, \Delta) = \lfloor \Delta \cdot \tau^{-1}(z) \rfloor \in R,$$

$$\text{Decode}(\mu, \Delta) = \Delta^{-1} \tau(\mu) \in \mathbb{C}^{n/2}.$$

- CKKS, Section 3 takes both maps over unchanged and cites Lecture 4, Section 7 for them.
- The $n/2$ coordinates of $\tau(a)$ are the CKKS slots, the complex analogue of Section 5's; since τ is multiplicative, ring multiplication is slotwise multiplication.
- The choice of $\langle 5 \rangle$ is what makes σ_5 of Section 6 a cyclic rotation of them, and σ_{-1} slotwise conjugation.
- Lecture 5, Section 8 reuses σ differently: $\sigma(R)$ is a *lattice*, and Ring-LWE hardness is stated in its geometry.

Example — encoding a complex vector

Take $n = 4$, $\Delta = 10^2$, $\zeta = \exp(\pi i/4)$, and

$$z = (1 + 4i, 5 - 2i) \in \mathbb{C}^2.$$

The evaluation matrix $A_{k,j} = \zeta^{kj}$, $k \in \{1, 3, 5, 7\}$, satisfies $A^*A = 4I$, so $(\Delta\tau^{-1}(z))_j = \frac{1}{4} \sum_k \zeta^{-kj} v_k$ with $v_1 = \Delta z_1$, $v_5 = \Delta z_2$, $v_7 = \overline{v_1}$, $v_3 = \overline{v_5}$.

Compute $\Delta\tau^{-1}(z)$, check that all four entries are real, and round to get μ .

Example — decoding, and the error

Same $n = 4$, $\Delta = 10^2$, and the μ just computed.

Using $\zeta^2 = i$, $\zeta^5 = -\zeta$ and $\zeta^{15} = \zeta^7 = \bar{\zeta}$, evaluate $\mu(\zeta)$ and $\mu(\zeta^5)$.

Divide by Δ to recover the slots, and compare with the true $z = (1 + 4i, 5 - 2i)$: by how much is each slot off?

Summary — the modulus, the switch, the digits, the degree

- 1 RNS is CRT coefficientwise: $R_Q \cong \prod_i R_{q_i}$, with chain $q_\ell = p_0 \cdots p_\ell$ and level ℓ . Dropping a level deletes one component.
- 2 Modulus switching sends $ct \mapsto \lfloor (q'/q)ct \rfloor$ in the class modulo t , preserving the message and dividing the noise by q/q' , subject to $q \equiv q' \pmod{t}$. CKKS's RESCALE is the same formula without the congruence.
- 3 $\langle g, G^{-1}(a) \rangle = a$ with $\|G^{-1}(a)\|_\infty \leq B/2$ *regardless of* a .
- 4 Read ct as $c_0 + c_1 S$; multiplication gives a quadratic in S , and relinearisation contracts $G^{-1}(c_3^*)$ against an evaluation key encrypting $g \cdot SK^2$, at the cost of $\sum_i c_{3,i}^* e_i$.

- 5 For $t \equiv 1 \pmod{2n}$, $R_t \cong \mathbb{F}_t^n$: n SIMD slots with componentwise $+$ and $\times$, computed by the NTT.
- 6 The $\sigma_k : X \mapsto X^k$, k odd, form $\mathbb{Z}_{2n}^\times = \langle 5 \rangle \times \langle -1 \rangle$ and permute slots by $j \mapsto jk$ — but change the key to $\sigma_k(SK)$, so each needs a key switch.
- 7 The canonical embedding evaluates at the complex roots of $X^n + 1$ and reduces to $\tau : K \rightarrow \mathbb{C}^{n/2}$. CKKS encodes by $\lfloor \Delta \tau^{-1}(z) \rfloor$ and decodes by $\Delta^{-1} \tau(\cdot)$.

Lecture 5 — Lattices and Hardness

Everything about *hardness* has been deferred. Why $As + e$ should look uniform, how Ring-LWE reduces to a worst-case lattice problem, and what the approximation factor is.

The canonical embedding returns there as geometry: $\sigma(R)$ is a lattice, and that is where the noise bounds of this lecture acquire a meaning.